THE PULLBACK OF A MODULE OF FINITE TYPE IS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The reduction of the statement to a module generated by finitely many global sections. Obtained by the same argument as in the proof of the author lemma lemma-pullback-locally-generated, invoked at source lines 1045-1047. Replaces local generation by an actual surjection from a finite direct sum of structure sheaves.
- The compatibility of the pullback with colimits and its right exactness. Invoked inside the counted proof at source lines 1048-1049 as a pointer to a lemma of this chapter. Transports the surjection from the finite direct sum to the pullback.

Source reference. Original lemma lemma-pullback-locally-generated: sources/stacks-modules/pinned-original-source.tex, source lines 1045–1045. Navigation pointer only: the counted text refers to the Stacks item lemma-pullback-locally-generated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-exactness-pushforward-pullback: sources/stacks-modules/pinned-original-source.tex, source lines 990–990. Navigation pointer only: the counted text refers to the Stacks item lemma-exactness-pushforward-pullback, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The pullback $f^*\mathcal{G}$ of a finite type $\mathcal{O}_Y$-module is a finite type $\mathcal{O}_X$-module.*

Proof. Arguing as in the proof of Lemma lemma-pullback-locally-generated we may assume $\mathcal{G}$ is globally generated by finitely many sections. We have seen that f^* commutes with all colimits, and is right exact, see Lemma lemma-exactness-pushforward-pullback. Thus if we have a surjection

$$\bigoplus\nolimits_{i=1,\ldots,n} \mathcal{O}_Y \to \mathcal{G} \to 0$$

then upon applying f^* we obtain the surjection

$$\bigoplus\nolimits_{i=1,\ldots,n} \mathcal{O}_X \to f^*\mathcal{G} \to 0.$$

This implies the lemma. □